EE502 - Linear Systems Theory II **Spring 2023**

Lecture 11

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11.1 Observability in CT-LTI Systems

In the context of observability of dynamical systems, it turns out that it is more convenient to think in terms of "unobservable states" and then connect it to the concept of observability and fully observable systems, as reflected in the following definitions.

Definition: For an LTI continuous-time state-space representation

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned}$$

A state x_u is said to be unobservable over $t \in [0, T)$, if with $x(0) = x_u$ and $\forall\, u(t)$ we get the same $y(t)$ as we would with $x(0) = 0$.

The set, $\bar{\mathcal{O}}_T$, of all unobservable states over $t \in [0, T)$ forms a vector space, $\bar{\mathcal{O}}_T \subset \mathbb{R}^n$, and the system is called fully observable over $t \in [0, T)$, if $\dim[\bar{\mathcal{O}}_T] = 0$.

Note that for linear dynamical systems, the observability of a state and system is independent of $u(t)$; in that respect, we will only analyze the zero-input response of the system in our derivations.

Theorem: The following statements regarding an initial state x_u are equivalent:

1. $x_u \in \bar{\mathcal{O}}_T$ (Unobservable over $t \in [0, T)$)
2. $x_u \in \bar{\mathcal{O}}_\tau$ for all $\tau > 0$
3. $x_u \in \mathcal{N}[\mathbf{O}]$, where the observability matrix is

$$\mathbf{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

Proof:
Step 1: (3) $\implies$ (2).
Let $x_u \in \mathcal{N}[\mathbf{O}]$, then

$$\mathbf{O}x_u = 0 \implies Cx_u = 0,\ CAx_u = 0,\ \ldots\ , CA^{n-1}x_u = 0$$

By the Cayley-Hamilton theorem, we can conclude that $CA^l x_u = 0$ for all integers $l \geq 0$. Evaluating the zero-input output response of the system with $x(0) = x_u$:

$$y_{zi}(\tau) = Ce^{A\tau}x_u = \sum_{l=0}^{\infty} \frac{\tau^l}{l!} CA^l x_u = 0$$

This holds for all $\tau > 0$, implying $x_u \in \bar{\mathcal{O}}_\tau$.

Step 2: (2) $\implies$ (1).
This is trivially true. If the state is unobservable for all $\tau > 0$, it is certainly unobservable for any specific $T > 0$.

Step 3: (1) $\implies$ (3).
If $x_u \in \bar{\mathcal{O}}_T$, then the zero-input output response is identically zero over the interval:

$$y_{zi}(t) = Ce^{At}x_u = 0, \quad \forall t \in [0, T]$$

Taking successive derivatives with respect to t and evaluating at $t = 0$ yields:

$$\begin{aligned} y_{zi}(0) &= 0 \implies Cx_u = 0 \\ \left[\frac{d}{dt}y_{zi}(t)\right]_{t=0} &= 0 \implies CAx_u = 0 \\ &\vdots \\ \left[\frac{d^{n-1}}{dt^{n-1}}y_{zi}(t)\right]_{t=0} &= 0 \implies CA^{n-1}x_u = 0 \end{aligned}$$

This means $\mathbf{O}x_u = 0$, or equivalently, $x_u \in \mathcal{N}[\mathbf{O}]$.

Similar to reachability, we show that for CT-LTI systems, observability and the unobservable (and observable) subspace are independent of time.

Ex 11.1 ***Problem:*** *Consider the following sparse 3×3 system:*

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 2 & -1 \end{bmatrix} \quad , \quad C = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

Determine if the system is fully observable. If not, find the unobservable subspace.

Solution: *Let's construct the observability matrix $\mathbf{O}$ to check the observability rank condition:*

$$\mathbf{O} = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 2 & -1 \end{bmatrix}$$

We can observe that the first column is entirely zero, while the second and third columns are linearly independent. Thus, $\text{rank}(\mathbf{O}) = 2 < 3$, meaning the system is not fully observable. The unobservable subspace is simply the null space of $\mathbf{O}$:

$$\bar{\mathcal{O}} = \mathcal{N}[\mathbf{O}] = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

This tells us that the initial condition of the first state variable, $x_1(0)$, is completely hidden and has absolutely no effect on the output $y(t)$ for any $t \geq 0$.

11.1.1 CT Observability Grammian

For a CT-LTI system, the observability Grammian over the interval $[0, t]$ is defined as

$$\mathbf{Q}(t) = \int_0^t e^{A^T\tau} C^T C e^{A\tau} d\tau$$

Theorem: $\mathcal{N}[\mathbf{Q}(t)] = \mathcal{N}[\mathbf{O}]\ \forall t > 0$.

Proof:
Part 1: Show $\mathcal{N}[\mathbf{O}] \subset \mathcal{N}[\mathbf{Q}(t)]$ for all $t > 0$.
If $x_u \in \mathcal{N}[\mathbf{O}]$, then we know from our previous theorem that $Ce^{A\tau}x_u = 0$ for all $\tau \geq 0$. Let's analyze if x_u is in the null-space of $\mathbf{Q}(t)$:

$$\mathbf{Q}(t)x_u = \int_0^t e^{A^T\tau} C^T \underbrace{(Ce^{A\tau}x_u)}_{0} d\tau = 0 \implies x_u \in \mathcal{N}[\mathbf{Q}(t)]$$

Part 2: Show $\mathcal{N}[\mathbf{Q}(t)] \subset \mathcal{N}[\mathbf{O}]$ for all $t > 0$.
Let $x_u \in \mathcal{N}[\mathbf{Q}(t)]$, which implies $\mathbf{Q}(t)x_u = 0$. Pre-multiplying by x_u^T gives:

$$x_u^T \mathbf{Q}(t) x_u = 0 \iff \int_0^t x_u^T e^{A^T\tau} C^T C e^{A\tau} x_u d\tau = \int_0^t \|Ce^{A\tau}x_u\|_2^2 d\tau = 0$$

Since the integrand is the squared Euclidean norm (which is strictly non-negative) and is continuous, the integral can only evaluate to zero if the integrand is identically zero:

$$Ce^{A\tau}x_u = 0 \quad \forall \tau \in [0, t]$$

Taking successive derivatives with respect to τ and evaluating at $\tau = 0$:

$$\begin{aligned}
[Ce^{A\tau}x_u]_{\tau=0} &= 0 \implies Cx_u = 0 \\
\left[\frac{d}{d\tau}(Ce^{A\tau}x_u)\right]_{\tau=0} &= 0 \implies CAx_u = 0 \\
&\vdots \\
\left[\frac{d^{n-1}}{d\tau^{n-1}}(Ce^{A\tau}x_u)\right]_{\tau=0} &= 0 \implies CA^{n-1}x_u = 0
\end{aligned}$$

Thus, $\mathbf{O}x_u = 0 \implies x_u \in \mathcal{N}[\mathbf{O}]$. We conclude $\mathcal{N}[\mathbf{Q}(t)] \subset \mathcal{N}[\mathbf{O}]$.

Ex 11.2 *Show that if $\dot{x} = Ax$ is asymptotically stable, then the observability Grammian at $t \to \infty$, $Q := \mathbf{Q}_\infty$, satisfies the following Lyapunov equation*

$$A^T Q + QA = -C^T C$$

, and this Lyapunov equation has a (unique) positive-definite solution for Q, if and only if, (A, C) is fully observable.

11.1.2 Further Results in CT-LTI Observability

The powerful Principle of Duality states that: **The pair (A, C) is observable if and only if the pair (A^T, C^T) is reachable.**

In view of this duality, we can use our reachability results to immediately derive various conclusions, tests, standard and canonical forms, etc., for observable and unobservable systems. The reader is highly encouraged to derive the following results, theorems, and claims by referring to the dual results in the reachability lecture.

Result 1: The unobservable sub-space, $\mathcal{N}[\mathbf{O}]$ is A invariant, i.e. $x \in \mathcal{N}[\mathbf{O}] \;\Rightarrow\; Ax \in \mathcal{N}[\mathbf{O}]$.

Result 2 (Modal Test for Unobservability): The pair (A, C) is *unobservable* $\iff$ there exists a right eigenvector $v \neq 0$ of A (where $Av = \lambda v$) such that $Cv = 0$. In this case, λ is called an unobservable mode.

Result 3 (PBH Rank Test for Observability): The pair (A, C) is *fully observable* $\iff$ the Popov-Belevitch-Hautus (PBH) observability matrix has full column rank for all complex numbers:

$$\operatorname{rank}\left[\begin{array}{c} \lambda I - A \\ \hline C \end{array}\right] = n, \;\forall \lambda \in \mathbb{C}$$

Result 4: Observability is invariant under state/similarity transformation.

11.1.3 Standard Form of Unobservable Systems

Let $\dot{x} = Ax + Bu$, $y = Cx + Du$ be a system that is not fully observable; then it is convenient and practical to choose coordinates (via similarity transformation) to highlight the unobservable and observable "spaces". Let

$$\begin{aligned} &\dim(\bar{\mathcal{O}}) = \bar{o} \;\&\; T = \left[\begin{array}{c|c} T_1 & T_2 \end{array}\right] \text{ where } T_2 \in \mathbb{R}^{n \times \bar{o}} \;, T_1 \in \mathbb{R}^{n \times (n-\bar{o})} \;, \\ &\bar{A} = T^{-1}AT \;,\; \bar{C} = CT \end{aligned}$$

Let's choose T_2 such that $\mathrm{Ra}(T_2) = \mathcal{N}(\mathbf{O}) = \bar{\mathcal{O}}$. In other words, the columns of T_2 span the null-space of the observability matrix (the unobservable subspace). Similarly, let's choose T_1 such that the columns of T are linearly independent. Let's analyze the similarity transformation on A:

$$\begin{aligned} &AT = A\left[\begin{array}{c|c} T_1 & T_2 \end{array}\right] = T\bar{A} = \left[\begin{array}{c|c} T_1 & T_2 \end{array}\right]\left[\begin{array}{c|c} A_{11} & A_{12} \\ \hline A_{21} & A_{22} \end{array}\right] \\ \Longrightarrow\; &AT_2 = T_1A_{12} + T_2A_{22} \end{aligned}$$

Note that since the unobservable sub-space is A-invariant, AT_2 must remain in $\mathrm{Ra}(T_2) = \mathcal{N}(\mathbf{O}) = \bar{\mathcal{O}}$. Because the columns of T_1 are linearly independent from T_2, for $T_1A_{12} + T_2A_{22}$ to remain entirely in $\mathrm{Ra}(T_2)$, the component along T_1 must be zero. Therefore, we must have $A_{12} = \mathbf{0}$.

Now let's analyze the effect of the transformation on the output matrix:

$$\bar{C} = CT = C\left[\begin{array}{c|c} T_1 & T_2 \end{array}\right] = \left[\begin{array}{c|c} CT_1 & CT_2 \end{array}\right] = \left[\begin{array}{c|c} C_1 & C_2 \end{array}\right]$$

Because T_2 spans the null-space of the observability matrix $\mathbf{O}$, and C is simply the first block-row of $\mathbf{O}$, any vector in $\mathrm{Ra}(T_2)$ is trivially in the null-space of C. Thus:

$$C_2 = CT_2 = \mathbf{0}$$

Therefore, the standard unobservable form takes the structure:

$$\begin{aligned} \dot{\bar{x}} &= \left[\begin{array}{c|c} A_{11} & \mathbf{0} \\ \hline A_{21} & A_{22} \end{array}\right] \bar{x} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} u \\ y &= \left[\begin{array}{c|c} C_1 & \mathbf{0} \end{array}\right] \bar{x} + Du \end{aligned}$$

Notice that the eigenvalues of A_{22} govern the unobservable modes of the system. These modes are completely "hidden" from the output $y(t)$, meaning their dynamic behavior does not affect the measurements at all. We will formalize this decomposition and structure further when we cover the Kalman Decomposition in an upcoming lecture.

Ex 11.3 ***Problem:*** *Consider a sparse 3×3 system that is already in the standard unobservable form:*

$$\dot{x} = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 0 \\ 1 & 0 & -3 \end{bmatrix} x \quad , \quad y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x$$

(a) Determine the unobservable subspace.
(b) Identify the unobservable mode(s).
(c) Demonstrate that an initial condition inside the unobservable subspace yields identically zero output despite internal state evolution.

Solution:
(a) Unobservable Subspace: *By partitioning, we have the observable part $A_{11} = -1$, the coupling block $A_{21} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and the unobservable block $A_{22} = \begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix}$. The observability matrix is:*

$$\mathbf{O} = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

We can clearly see $\mathrm{rank}(\mathbf{O}) = 1 < 3$. The unobservable subspace is the null-space of $\mathbf{O}$:

$$\bar{\mathcal{O}} = \mathcal{N}[\mathbf{O}] = \mathrm{span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

(b) Unobservable Modes: *The unobservable block A_{22} is diagonal and has eigenvalues $\lambda_2 = -2$ and $\lambda_3 = -3$. The corresponding right eigenvectors for the full matrix A are simply $v_2 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T$ and $v_3 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T$. Because $v_2, v_3 \in \mathcal{N}[\mathbf{O}]$, we get $Cv_2 = 0$ and $Cv_3 = 0$. The modal responses e^{-2t} and e^{-3t} are completely unobservable.*

(c) Zero-Output Demonstration: *If we start the system with an initial condition strictly in the unobservable subspace, say $x(0) = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, we can calculate the state trajectory $x(t)$ without needing to compute the full matrix exponential e^{At}.*

Since the first equation is $\dot{x}_1(t) = -x_1(t)$ and we start with $x_1(0) = 0$, we know $x_1(t) = 0$ for all $t \geq 0$. Substituting $x_1(t) = 0$ into the remaining rows completely decouples the ODEs:

$$\begin{aligned} \dot{x}_2(t) &= 0 - 2x_2(t) \quad \Rightarrow \quad x_2(t) = e^{-2t} x_2(0) = e^{-2t} \\ \dot{x}_3(t) &= 0 - 3x_3(t) \quad \Rightarrow \quad x_3(t) = e^{-3t} x_3(0) = e^{-3t} \end{aligned}$$

The internal state evolution is therefore:

$$x(t) = \begin{bmatrix} 0 \\ e^{-2t} \\ e^{-3t} \end{bmatrix}$$

However, the measured zero-input output remains identically zero:

$$y(t) = Cx(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ e^{-2t} \\ e^{-3t} \end{bmatrix} = 0 \quad \textit{for all } t \geq 0$$

The dynamic behavior of these hidden modal states cannot be seen from the output.

11.2 Observability in DT-LTI Systems

Definition: For an LTI discrete-time state-space representation

$$\begin{aligned} x[k+1] &= A[k]x + B[k]u \\ y[k] &= C[k]x + D[k]u \end{aligned}$$

A state x_u is said to be m-step unobservable if with $x[0] = x_u$ and $\forall\, u[k]\, k \in [0, m)$ we get the same $y[k]\, k \in [0, m)$ as we would with $x[0] = 0$.

The set, $\bar{\mathcal{O}}_m$, of all m-step unobservable states forms a vector space, $\bar{\mathcal{O}}_m \subset \mathbb{R}^n$.

Similar to the CT-LTI case, we will only analyze the zero-input response of the system in our observability-related derivations.

If $x_u \in \bar{\mathcal{O}}_m$, then the zero-input response is exactly zero for the first m steps:

$$y[k] = CA^k x_u = 0\ ,\ \forall k \in [0, m)$$

Expanding this condition step-by-step yields:

$$\begin{aligned} k = 0 &\implies Cx_u = 0 \\ k = 1 &\implies CAx_u = 0 \\ &\ \ \vdots \\ k = m-1 &\implies CA^{m-1}x_u = 0 \end{aligned} \qquad \Rightarrow \qquad \underbrace{\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{m-1} \end{bmatrix}}_{\mathbf{O}_m} x_u = 0$$

This implies that x_u is m-step unobservable if and only if $x_u \in \mathcal{N}[\mathbf{O}_m]$.

Theorem: $\mathcal{N}[\mathbf{O}_i] = \mathcal{N}[\mathbf{O}_n] \subset \mathcal{N}[\mathbf{O}_m]$ for $m < n < i$.

Corollary: If x_u is n-step unobservable, then it is i-step unobservable $\forall i \geq n$.

In this context for DT systems, the unobservable space is defined as $\bar{\mathcal{O}} = \mathcal{N}[\mathbf{O}_n]$ and the system is fully observable if and only if

$$\bar{\mathcal{O}} = \{0\} \iff \dim(\mathcal{N}[\mathbf{O}_n]) = 0 \iff \text{rank}[\mathbf{O}_n] = n$$

11.2.1 DT Observability Grammian

m-step observability Grammian is defined as

$$Q_m = \mathbf{O}_m^T \mathbf{O}_m = \sum_{i=0}^{m-1} (A^i)^T C^T C A^i$$

Theorem: $\mathcal{N}[\mathbf{O}_m] = \mathcal{N}[\mathbf{Q}_m]$.

Let $x[k+1] = Ax[k]$ be asymptotically stable, then $Q = \mathbf{Q}_\infty$ is well defined and satisfies the Lyapunov equation below

$$A^T Q A - Q = -C^T C$$

and this Lyapunov equation has a (unique) positive-definite solution for Q, if and only if, (A, C) is fully observable.

Corollary: Total Observability Theorem
The following statements are mathematically equivalent for a discrete-time system:

- The system is fully observable.
- The pair (A, C) is observable.
- The observability matrix has full rank: $\text{rank}[\mathbf{O}_n] = n$.
- **Modal Test:** $Cv \neq 0$ for all right eigenvectors v of A (i.e., $\forall v \neq 0$ such that $Av = \lambda v$).
- **PBH Test:** rank $\left[\begin{array}{c} \lambda I - A \\ \hline C \end{array}\right] = n, \ \forall \lambda \in \mathbb{C}$.